

JDAKS Infra

Road Project Management System

Laravel Web Application

Full-Stack Development Plan

Tech Stack	Laravel 11 + MySQL + Bootstrap 5
Authentication	Admin-only login, RBAC, Password Reset
Mobile Friendly	Fully responsive Bootstrap 5 UI
Estimated Timeline	14 Weeks (3.5 Months)
Prepared	May 2026

1. Executive Summary

This document is a comprehensive full-stack development plan for migrating the existing JDAKS Infra single-page HTML dashboard into a production-grade, multi-user Laravel 11 web application. The system currently operates as a standalone browser file using localStorage. The Laravel rebuild will provide secure server-side data persistence, role-based access, and a mobile-first UI that multiple team members can access simultaneously from any device.

Key objectives:

- Replace browser localStorage with a secure MySQL database
- Implement admin-controlled user management with Role-Based Access Control (RBAC)
- Preserve all existing module functionality: Progress Tracker, DSR Calculator, Billing & RA Bills, Inventory, Price Variation, and Expense Sheet
- Build a fully responsive Bootstrap 5 interface optimised for both desktop and mobile
- Provide a secure login system with forgot-password email flow (no self-registration)
- Support PDF export, Excel export, and cloud sync equivalent via server-side APIs

2. System Overview & Module Analysis

The existing dashboard is a fully functional Road Project Management System built for JDAKS Infra. Six primary modules have been identified from the codebase:

#	Module	Description	Key Features
1	Progress Tracker	Log daily work quantities against BOQ items, track chainage, date-wise log, and road overview	BOQ items, chainage map, status tracking, PDF reports
2	DSR Calculator	Analyse cost per MT/CUM for Bituminous, Base Course, Concrete, Earthwork, and Other items. Compares rate against Net BOQ rate.	Material, Labour, Machinery breakdown, bid % recommendation, deductions analysis
3	Billing & RA Bills	Raise Running Account bills, apply statutory deductions (SD, IT TDS, Labour Cess, GST TDS), and track payment status	LOA configuration, bill lifecycle (draft/submitted/certified/approved/paid), deduction tracking
4	Inventory & Materials	Track bitumen and cement stock, log consumption against progress entries, monitor procurement gaps	Stock ledger, purchases, consumption, wastage, low-stock alerts, reorder thresholds
5	Price Variation (PV)	PV Calculator with component indices (Labour, Cement, Steel, Bitumen, POL, Other, Plant), month-wise PVC, bitumen P&L	Index-based PVC calculations, month-wise comparison, bitumen profit & loss
6	Expense Sheet	Daily labour, machinery & other costs with net margin vs earned value analysis	Expense categories, date-wise logging, margin analysis vs BOQ earned

In addition to these six core modules, the application includes a Project Hub (central dashboard per project), a global Home screen listing all projects with summary cards, and Settings/configuration screens.

3. Technology Stack

3.1 Back-End

- PHP 8.2+ with Laravel 11 (LTS) framework
- MySQL 8.0 database with Eloquent ORM
- Laravel Sanctum for API token authentication (for future mobile apps)
- Laravel Queues (Redis or database driver) for background jobs — PDF generation, email
- Laravel Mail with SMTP/Mailgun for password reset emails
- Spatie Laravel-Permission for Role-Based Access Control (RBAC)
- DomPDF / Laravel-DOMPDF for server-side PDF generation of RA bills and progress reports
- Maatwebsite/Excel (PhpSpreadsheet) for BOQ import/export and Excel generation

3.2 Front-End

- Bootstrap 5.3 for responsive, mobile-first grid and components
- Vanilla JavaScript + Alpine.js for lightweight reactivity (modals, tabs, dropdowns)
- Chart.js 4 for progress charts, billing charts, and analytics graphs
- Blade templating engine with reusable components and layouts
- Laravel Vite for asset bundling (CSS + JS compilation)

3.3 Infrastructure

- Apache / Nginx web server
- MySQL 8.0 relational database
- Redis (optional) for session storage and queue driver
- Laravel Telescope (development) for debugging and query inspection
- Laravel Horizon (optional) for queue monitoring

4. Authentication & User Management

4.1 Login System Rules

Per requirements, the system follows strict authentication rules:

- No public registration — the login page contains only Sign In and Forgot Password
- Only the Admin (seeded on first deployment) can create new user accounts (Site Manager or Field Engineer)
- Forgot Password sends a time-limited reset link via email (Laravel built-in Password Broker)
- Failed login attempts are rate-limited: account locks after 5 failed attempts for 15 minutes
- Sessions expire after 2 hours of inactivity
- Passwords must meet minimum complexity (8+ characters, mixed case, number)

4.2 Roles & Permissions (RBAC)

Using Spatie Laravel-Permission, the following roles are defined. Admin can assign any role when creating a user account:

Role	Level	Permissions
Admin	3 — Full	All permissions. Manage users, create/edit/delete projects, all modules, system settings. Cannot be deleted. Only role that can create new user accounts.
Site Manager	2 — Medium	View all projects. Create and edit progress entries, expenses, inventory logs for assigned projects. View billing. Cannot delete projects or manage users.
Field Engineer	1 — Limited	Log daily progress entries and expenses for assigned projects. View-only on BOQ, Billing, and Calculator. Cannot delete or manage anything.

4.3 User Account Fields

- Name, Email (unique), Password (hashed bcrypt)
- Role (assigned by Admin — only Site Manager or Field Engineer selectable), Status (active/inactive)
- Project assignments (many-to-many: user can be assigned to multiple projects)
- Last login timestamp, Created by (admin who created the account)

4.4 Admin User Management Screen

- List all users in a table with name, email, role, status, last login, actions
- Create User modal: name, email, role dropdown (Site Manager / Field Engineer), auto-generate temporary password with copy button
- Edit User: change name, email, role, status (active/inactive)
- Reset Password: send a reset link to user email
- Assign Projects: multi-select project assignment per user
- Delete User (with confirmation — Admin account itself cannot be deleted)

5. Database Schema

The following tables are required. All tables include created_at, updated_at, and soft-delete (deleted_at) columns unless noted.

5.1 Core Tables

Table	Key Columns	Notes
users	id, name, email, password, role_id, status, created_by, last_login_at	Spatie permission integration
roles	id, name, guard_name	Spatie: admin, site_manager, field_engineer
permissions	id, name, guard_name	Granular permissions per module (view_billing, edit_progress, etc.)
model_has_roles	role_id, model_id, model_type	Pivot: user-to-role assignments
project_user	project_id, user_id, assigned_by, assigned_at	Pivot: project access assignments

5.2 Project Tables

Table	Key Columns	Notes
projects	id, name, package, location, contractor, chainage_start, chainage_end, start_date, end_date, deadline, bid_date, bid_type, bid_pct, bit_grade, dual_lane, notes, created_by	Central project record
boq_items	id, project_id, category, description, unit, quantity, rate, amount, sort_order	Bill of Quantities items per project
progress_entries	id, project_id, boq_item_id, date, quantity, chainage_from, chainage_to, status, notes, created_by	Daily progress log entries
billing_configs	id, project_id, loa_number, tendered_value, bid_type, bid_pct, sd_pct, it_tds_pct, labour_cess_pct, gst_tds_pct, configured_at	Billing configuration per project
ra_bills	id, project_id, bill_number, bill_date, gross_amount, deductions_json, net_payable, status, submitted_at, certified_at, paid_at, paid_amount, notes	Running Account bills lifecycle

5.3 Module Tables

Table	Key Columns	Notes
dsr_items	id, project_id, boq_item_id, material_json, labour_json, machinery_json, deduction_pct_json, updated_by	DSR Calculator data per BOQ item

inventory_materials	id, project_id, material_code, name, unit, reorder_qty, opening_stock	Master material catalogue per project
inventory_transactions	id, project_id, material_id, txn_type (purchase/consumption/wastage), date, quantity, rate, amount, reference_id, notes, created_by	Stock ledger entries
pv_indices	id, project_id, month, labour_idx, cement_idx, steel_idx, bitumen_idx, pol_idx, other_idx, plant_idx, base_month, created_by	Monthly price variation index entries
expenses	id, project_id, date, category (labour/machinery/other), description, amount, created_by	Daily project expense entries

6. Application Architecture

6.1 Laravel MVC Structure

The application follows standard Laravel MVC conventions with the following structure:

- app/Http/Controllers/Auth/ — LoginController, ForgotPasswordController, ResetPasswordController
- app/Http/Controllers/Admin/ — UserController, RoleController (admin-only)
- app/Http/Controllers/ — ProjectController, BoqItemController, ProgressController, BillingController, DsrController, InventoryController, PvController, ExpenseController
- app/Http/Middleware/ — RoleMiddleware, ProjectAccessMiddleware
- app/Models/ — User, Project, BoqItem, ProgressEntry, RaBill, BillingConfig, DsrItem, InventoryMaterial, InventoryTransaction, PvIndex, Expense
- app/Policies/ — ProjectPolicy, BillingPolicy, ProgressPolicy (Laravel Gates & Policies)
- app/Services/ — BillingService, ProgressService, PvCalculationService, PdfExportService, ExcelExportService
- resources/views/layouts/ — app.blade.php (authenticated layout with sidebar+topbar), auth.blade.php (login layout)
- resources/views/ — auth/, admin/, projects/, progress/, billing/, dsr/, inventory/, pv/, expenses/

6.2 Routing Groups

Routes are organised into three groups:

- Guest routes (no auth): /login, /forgot-password, /reset-password/{token}
- Authenticated routes (all logged-in users): /dashboard, /projects/*, all module routes
- Admin-only routes (admin middleware): /admin/users/*, /admin/settings — accessible only by the Admin role

6.3 Middleware Stack

- auth — Laravel standard authentication check
- verified — email verification (optional, can disable for internal tool)
- role:admin — restrict to Admin user only
- ProjectAccess — verify the authenticated user is assigned to the requested project
- ModulePermission — verify user role has access to the specific module being accessed

7. Screen-by-Screen Feature Specification

7.1 Login Screen

- Clean centred card layout (matching existing HTML design) — JDAKS Infra logo, company name, subtitle
- Email and password fields with show/hide password toggle
- Sign In button — rate-limited, lockout after 5 failed attempts
- Forgot Password link — opens /forgot-password page
- No Register link — registration is admin-only via back-office
- Error messages: invalid credentials, account locked (with countdown), account inactive

7.2 Forgot Password Flow

- User enters email address on /forgot-password page
- Laravel sends a signed time-limited reset link to the registered email
- User clicks link, enters new password + confirmation on /reset-password/{token}
- On success, redirect to login with success message

7.3 Dashboard — Project List (Home Screen)

- Responsive card grid showing all projects the user has access to
- Each card: project name, package, location, BOQ value, % complete progress bar, status pills
- Search/filter bar: search by name, filter by status (Active / Completed / All)
- New Project button (visible to Admin only)
- Quick stats strip at top: total projects, total BOQ value, total earned

7.4 Project Hub

- Project summary strip: name, package, location, contractor, BOQ value, earned, % complete, chainage progress bar
- Bitumen Grade selector (VG-30 / VG-40) — saved per project
- Dual Lane toggle (On / Off) — saved per project
- Bid % configuration: Below / At Par / Above with percentage input
- Six module cards: Progress Tracker, DSR Calculator, Billing & RA Bills, Inventory & Materials, Price Variation, Expense Sheet
- Each module card shows a live badge summary (e.g., items completed, bills pending, stock status)
- Edit Project and Delete Project buttons (Admin only)

7.5 Progress Tracker

- Tab bar: BOQ Items | Chainage Map | Date Log | Road Overview
- BOQ Items tab: list of all BOQ items with category colour coding, quantity done/total, status pill (Not Started / In Progress / Completed), expandable detail row for logging entries
- Add Entry modal: date, chainage from/to, quantity, status (Completed / Partial), notes
- Filter by category (Earthwork / Bituminous / Base Course / Concrete / Drainage / Signage / Furniture)
- Chainage Map tab: visual strip showing progress along road length with colour-coded segments
- Date Log tab: date-wise grouped view of all entries, collapsible cards

- Road Overview tab: summary statistics with Chart.js progress charts
- Export to PDF and Export to Excel buttons

7.6 DSR Calculator

- PIN-protected access (configurable per-role in admin settings)
- Item list with expandable DSR analysis rows
- Each item: Material sub-table, Labour sub-table, Machinery sub-table (add/edit/delete rows)
- Auto-calculated totals: cost per unit, earned rate vs tendered rate, margin %
- Deductions configuration panel (SD %, IT TDS %, Labour Cess %, GST TDS %)
- Cost Analysis panel: Theoretical vs Actual vs Variance (chainage-based)
- Bid % recommendation based on deduction percentages

7.7 Billing & RA Bills

- Configuration step: LOA number, tendered value, bid type/%, deduction percentages
- RA Bill list: bill number, date, gross amount, net payable, status badge, action buttons
- Create/Edit RA Bill: select BOQ items and quantities, auto-calculate gross, apply deductions, compute net payable
- Bill status workflow: Draft > Submitted > Certified > Approved > Paid
- Mark as Paid: enter paid amount and date
- PDF export per bill (formal RA Bill format)
- Payment summary: total bills raised, total paid, total outstanding

7.8 Inventory & Materials

- Materials list with current stock, unit, reorder threshold, status (OK / Low / Out)
- Add Purchase: material, date, quantity, rate — auto-calculates amount, updates stock
- Log Consumption: material, date, quantity, linked to progress entry (optional)
- Log Wastage: material, date, quantity, reason
- Stock Ledger view: chronological transaction log with running balance
- Low stock alerts highlighted in red with reorder quantity suggestion

7.9 Price Variation Calculator

- Component indices input: Labour, Cement, Steel, Bitumen, POL, Other, Plant
- Base month and current month selection
- Auto-calculated PVC per component and total PVC amount
- Month-wise PVC table for multiple billing periods
- Bitumen P&L analysis: purchase price vs index-adjusted price

7.10 Expense Sheet

- Date-wise expense log with categories: Labour, Machinery, Other
- Add Expense: date, category, description, amount
- Summary: total expenses by category, total vs earned value, net margin calculation

- Chart: expenses trend vs earned value over time

7.11 Admin — User Management

- User list table: name, email, role badge, status, last login, assigned projects count, actions
- Create User drawer/modal: name, email, role selector, auto-generated password with copy button, send welcome email option
- Edit User: update name, email, role, status
- Assign Projects: multi-select checkbox list of projects per user
- Reset Password: trigger reset email
- Deactivate/Reactivate user (toggle status)
- Activity log: last login, account creation info

8. UI/UX Design Guidelines

8.1 Design System

The Bootstrap 5 UI should closely replicate the visual identity of the existing HTML dashboard:

- Colour palette: background #EFEFEF, surface white, text #1A1A1A, amber accent #C47C0A, green #1A7A3A, red #B03020, blue #1A4FA0
- Typography: system-ui / Segoe UI, 14px base, 11px labels, 22px metric values
- Border radius: 8px standard, 12px large cards
- Status pills: amber (in progress), green (completed), red (overdue/low stock), blue (submitted)
- Progress bars: 7px height, amber fill (< 100%), green fill (100%)

8.2 Mobile Responsiveness

- Navigation: top navbar collapses to hamburger menu on mobile; sidebar drawer for authenticated screens
- Grid: 4-column on desktop, 2-column on tablet, 1-column on mobile (Bootstrap col-md-*, col-sm-*)
- Tables: horizontal scroll wrapper on mobile (table-responsive), or card view transformation for key tables
- Modals: full-screen on mobile (modal-fullscreen-sm-down Bootstrap class)
- Buttons: full-width on mobile, minimum tap target 44px
- Sticky top navbar with project name and back navigation on all inner screens

8.3 Navigation Structure

- Global topbar: logo, project name breadcrumb, back button, user avatar dropdown (profile, logout)
- Side drawer (mobile) / left sidebar (desktop) for module navigation within a project
- Breadcrumbs: Home > Project Name > Module Name
- Tab bars within modules (e.g., Progress Tracker tabs: BOQ / Chainage / Date Log / Overview)

9. API & Export Features

9.1 Internal AJAX APIs

Laravel controllers will expose JSON endpoints (resource routes with Accept: application/json) for dynamic interactions:

- POST /projects/{id}/progress — add progress entry (returns updated BOQ item summary)
- GET /projects/{id}/boq/summary — returns BOQ completion JSON for dashboard cards
- POST /projects/{id}/billing/bills — create RA bill
- PATCH /projects/{id}/billing/bills/{billId}/status — update bill status
- POST /projects/{id}/inventory/transactions — log stock transaction
- GET /projects/{id}/inventory/stock — current stock positions JSON

9.2 PDF Export

- Progress Report PDF: project details, BOQ summary table, chainage map, date log (via DomPDF Blade view)
- RA Bill PDF: formal bill format with LOA details, item quantities, deductions, net payable, signature blocks
- Expense Report PDF: date-wise expenses, category totals, margin summary

9.3 Excel/CSV Export

- BOQ Export: full BOQ with quantities done, earned value, status
- Progress Log Export: date-wise entry log with all fields
- Billing Summary Export: all RA bills with payment status
- Inventory Ledger Export: full stock transaction history
- BOQ Import: Excel template import for bulk BOQ item creation (matching existing HTML import feature)

10. Development Phases & Timeline

Estimated for a team of 1 Full-Stack Developer. Add 40% buffer for QA and client review cycles.

#	Phase	Deliverables	Duration	Milestone
1	Project Setup & Architecture	Laravel 11 install, MySQL schema migrations, Spatie RBAC setup, Bootstrap 5 layout skeleton, CI/CD pipeline, .env configuration	1 week	Repo + DB schema ready
2	Authentication & User Management	Login page, forgot/reset password, admin user CRUD, role assignment, project-user assignment, middleware	1.5 weeks	Auth system live, admin can create users
3	Project Management Core	Project CRUD (create/edit/delete), BOQ item CRUD, BOQ Excel import, project list dashboard, project hub screen	2 weeks	Projects & BOQ fully functional
4	Progress Tracker	Progress entry CRUD, BOQ item status tracking, chainage logic, date log view, chainage map view, road overview with Chart.js, PDF & Excel export	2.5 weeks	Progress module complete
5	DSR Calculator	DSR item data entry (Material/Labour/Machinery), cost analysis panel, deductions config, bid % recommendation, PIN protection	1.5 weeks	DSR module complete
6	Billing & RA Bills	Billing config, RA bill creation with deductions, bill status workflow, payment tracking, PDF bill export	2 weeks	Billing module complete
7	Inventory & Materials	Material catalogue, purchase/consumption/wastage transactions, stock ledger, low-stock alerts	1.5 weeks	Inventory module complete
8	Price Variation & Expense Sheet	PV index entry, PVC calculation, month-wise table, expense entry CRUD, margin analysis	1.5 weeks	PV + Expense modules complete
9	Mobile Optimisation & Polish	Responsive testing all screens, mobile table cards, full-screen modals, touch-friendly UI, performance optimisation	1 week	Mobile-ready on all devices
10	Testing, UAT & Deployment	Unit tests, feature tests, UAT with client, bug fixes, production server setup, SSL, final deployment	1 week	Production live

Total Estimated Duration: 16 weeks (with QA buffer) | Minimum Fast Track: 14 weeks

11. Recommended Laravel File Structure

Key files and directories for the project:

- app/Http/Controllers/Auth/ — LoginController, ForgotPasswordController, ResetPasswordController
- app/Http/Controllers/Admin/ — UserController, SettingsController (Admin role only)
- app/Http/Controllers/ — ProjectController, BoqController, ProgressController
- app/Http/Controllers/ — BillingController, DsrController, InventoryController, PvController, ExpenseController
- app/Http/Middleware/ — RoleMiddleware, ProjectAccessMiddleware, ModulePermissionMiddleware
- app/Models/ — User, Project, BoqItem, ProgressEntry, RaBill, BillingConfig, DsrItem
- app/Models/ — InventoryMaterial, InventoryTransaction, PvIndex, Expense
- app/Policies/ — ProjectPolicy, BillingPolicy, ProgressPolicy
- app/Services/ — BillingService, ProgressService, PvCalculationService
- app/Services/ — PdfExportService, ExcelExportService
- database/migrations/ — one migration per table (20+ files)
- database/seeds/ — SuperAdminSeeder, RolePermissionSeeder, DemoDataSeeder
- resources/views/layouts/ — app.blade.php, auth.blade.php, admin.blade.php
- resources/views/auth/ — login.blade.php, forgot-password.blade.php, reset-password.blade.php
- resources/views/admin/ — users/ (index, create, edit, show)
- resources/views/projects/ — index, create, edit, hub
- resources/views/progress/ — index (with tabs for BOQ, chainage, date-log, overview)
- resources/views/billing/ — index, create, show (PDF view)
- resources/views/dsr/ — index
- resources/views/inventory/ — index
- resources/views/pv/ — index
- resources/views/expenses/ — index
- resources/views/pdf/ — progress-report.blade.php, ra-bill.blade.php
- resources/js/ — app.js, progress-tracker.js, dsr-calculator.js, charts.js
- resources/css/ — app.css (custom variables matching existing dashboard colours)
- routes/web.php — all web routes grouped by middleware
- routes/api.php — AJAX JSON endpoints
- tests/Feature/ — AuthTest, ProjectTest, ProgressTest, BillingTest, InventoryTest

12. Security Considerations

- CSRF protection on all forms (Laravel @csrf token, built-in)
- SQL injection prevention via Eloquent ORM and parameterised queries
- XSS prevention: all Blade output auto-escaped with {{ }}, raw output avoided
- Password hashing: bcrypt via Laravel Hash facade
- Rate limiting: login endpoint throttled (5 attempts / 15 min) via Laravel RateLimiter
- Authorisation: every controller action checks Policy or Gate before acting
- ProjectAccess middleware: prevents users accessing projects they are not assigned to
- Sensitive routes (Admin User Management) protected by role:admin middleware — only the Admin account can access
- Password reset tokens: single-use, expire in 60 minutes
- HTTPS enforced in production via ForceHttps middleware or server config
- Environment variables: all secrets in .env (never committed to version control)
- Session security: secure, httpOnly, sameSite=lax cookie flags

13. Testing Strategy

13.1 Automated Tests (PHPUnit / Pest)

- Feature tests for all authentication flows: login success/failure, logout, password reset
- Feature tests for RBAC: verify Admin can access all routes, Site Manager cannot access /admin/*, Field Engineer has view-only on billing/DSR
- Feature tests for CRUD operations: project, BOQ, progress entry, RA bill, inventory
- Unit tests for calculation services: BillingService, PvCalculationService, BOQ earned value
- Browser tests (Laravel Dusk) for critical UI flows: login, create project, add BOQ item, log progress

13.2 Manual / UAT Testing

- Mobile browser testing: Chrome/Safari on Android and iOS
- Responsive layout verification at 375px, 768px, 1024px, 1440px breakpoints
- PDF export validation: verify bill and progress report layout
- Excel import/export roundtrip: import BOQ, export, re-import
- Multi-user concurrent testing: two users editing different projects simultaneously
- Data migration test: import existing HTML dashboard JSON export into the new database

14. Data Migration from Existing HTML Dashboard

The existing dashboard stores all data in browser localStorage as JSON. A migration utility is needed to move this data to the new MySQL database:

1. Add a Data Export feature to the existing HTML dashboard: a button to download all localStorage data as a single JSON file (projects, BOQ, entries, billing, inventory, DSR, PV, expenses)
2. Build a Laravel Import Command (php artisan import:dashboard) that reads the exported JSON and inserts records into the correct MySQL tables
3. Map JSON fields to database columns (field name mapping documented in migration guide)
4. Handle deduplication: check for existing project names before inserting
5. Validate all chainage values, date formats, and numeric fields during import
6. Provide an import report: number of records imported per table, any skipped records with reasons
7. Admin can trigger import from the Admin panel (Upload JSON file > Preview > Confirm Import)

15. Deployment & Server Requirements

15.1 Server Requirements

- PHP 8.2+ with extensions: OpenSSL, PDO, Mbstring, Tokenizer, XML, Ctype, JSON, BCMath, Fileinfo
- MySQL 8.0+
- Composer 2.x
- Node.js 18+ and npm (for Vite asset compilation)
- Apache 2.4+ or Nginx 1.18+ with mod_rewrite / try_files
- SSL certificate (Let's Encrypt recommended)
- Minimum 1 GB RAM, 20 GB disk

15.2 Deployment Checklist

8. Clone repository to server, set correct file permissions (storage/, bootstrap/cache/)
9. Copy .env.example to .env, configure APP_KEY, DB credentials, MAIL settings
10. Run: composer install --optimize-autoloader --no-dev
11. Run: php artisan key:generate
12. Run: php artisan migrate --force
13. Run: php artisan db:seed --class=RolePermissionSeeder
14. Run: php artisan db:seed --class=AdminSeeder (creates the single Admin account)
15. Run: npm install && npm run build (Vite production build)
16. Run: php artisan config:cache && php artisan route:cache && php artisan view:cache
17. Configure web server document root to /public
18. Configure HTTPS redirect and HTTP to HTTPS 301
19. Test login, create project, and all module flows in production

15.3 Initial Admin Credentials

The AdminSeeder creates the single Admin account. Change the password immediately after first login:

- Email: admin@jdaksinfra.com
- Password: Auto-generated and displayed once in seeder output — save it immediately

16. Future Enhancements (Post v1.0)

- Mobile App: React Native or Flutter app using Laravel Sanctum API tokens
- Notifications: in-app and email notifications for bill status changes, low stock alerts, deadline reminders
- Audit Log: full activity trail showing who changed what and when, per project
- Multi-Company Support: multi-tenancy for managing multiple infrastructure companies
- Document Attachments: upload and attach PDF/image files to projects, bills, and entries
- Dashboard Analytics: company-wide analytics across all projects (total earned, utilisation rates, project health)
- Offline Mode: PWA (Progressive Web App) support for field engineers with offline data sync
- WhatsApp / SMS Alerts: notify site managers of bill approvals and low stock via messaging APIs
- Two-Factor Authentication (2FA): TOTP app (Google Authenticator) for admin users

Appendix A: Key .env Configuration

Essential environment variables to configure before deployment:

- APP_NAME=JDAKS_Infra
- APP_ENV=production
- APP_KEY= (generated by php artisan key:generate)
- APP_URL=https://yourdomain.com
- DB_CONNECTION=mysql, DB_HOST, DB_PORT=3306, DB_DATABASE, DB_USERNAME, DB_PASSWORD
- MAIL_MAILER=smtp (or mailgun/ses for production)
- MAIL_HOST, MAIL_PORT, MAIL_USERNAME, MAIL_PASSWORD, MAIL_FROM_ADDRESS
- SESSION_DRIVER=database (or redis for scale)
- SESSION_LIFETIME=120 (minutes)
- QUEUE_CONNECTION=database (or redis)

Appendix B: Composer Dependencies

- laravel/laravel: ^11.0 — core framework
- spatie/laravel-permission: ^6.0 — RBAC roles & permissions
- barryvdh/laravel-dompdf: ^2.0 — PDF generation
- maatwebsite/excel: ^3.1 — Excel import/export
- laravel/telescope: ^5.0 — development debugging (dev only)
- laravel/sanctum: ^4.0 — API token authentication